

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0178**

ASSESSMENT PATTERN: **COMP0178A7PE 001**

MODULE NAME : **Database Fundamentals**

LEVEL : **Postgraduate**

DATE: : **16/05/2023**

TIME : **10:00**

DURATION : **2 Hours**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

**Year
2022-23**

**EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL.
PLACE EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE
EXAMINATION ENVELOPE**

Hall instructions	N/A
Additional materials	N/A
Standard Calculators	Yes
Non-Standard Calculators	No

TURN OVER

Suitable for Cohort(s): 2022/23, 2021/22

There are FIVE questions in total. Answer ALL questions. Calculators are permitted.

A maximum of 100 marks is available. The marks available for each part of each question are indicated in square brackets [n].

- 1) [Database design] Is the following schema in 3rd normal form? Justify your answer: If it is in 3rd normal form, briefly explain how you know. If it is not in 3rd normal form, point out violation(s) and explain how to correct them.

Teachers (teacherID, fName, lName, email)

Students (studentID, fName, lName, year, email)

Module (moduleID, moduleCode, moduleName, subject, desc)

Class (classID, moduleID, term, time, listOfStudents)

Teaching (teacherID, classID, moduleCode, term, classroom)

Assume: A module describes a specific curriculum taught throughout the term. A single module is often split into multiple classes that each contain a subset of the students taking that module. All classes are taught by exactly one teacher and is scheduled at the same time every day in the same classroom. A student is only enrolled in one class per module although they take multiple modules in a single term. Unlike students, teachers can teach the same module to multiple classes/groups of students in the same week.

[Total for Question 1: 15 marks]

- 2) [Query processing] Consider the following database schema used to support a real estate agency:

Staff (staffNo, fName, lName, position, sex, DOB, salary, branchNo);

Branch (branchNo, address, city);

Clients (clientNo, fName, lName, title, address, telNo, prefType);

Viewing (clientNo, propertyNo, viewdate, comment);

Property (propertyNo, address, type, rooms, rent, staffNo);

[Question 2 continues over page]

[Question 2 continued]

- (a) Write a SQL query to answer a question that the real estate agency might have. An example of such a question would be the names of all clients who have visited properties with 3 or more rooms. Your query should require at least three tables.

[5 marks]

- (b) Translate the SQL query you wrote in the previous part to a relational algebra expression that accomplishes the same task.

[5 marks]

- (c) Translate the relational algebra expression you gave above into (1) a relational algebra tree that matches exactly the expression, and (2) a second relational algebra tree that is different from (not identical to) but equivalent in meaning to the first.

[10 marks]

- (d) For the relational algebra trees you've given, estimate the number of disk reads that would be required under the following assumptions: No indexes exist. Tuples are read one at a time (1 tuple = 1 disk read). All tables fit entirely in memory. All intermediate calculations are stored to disk. However, the final result is returned directly to the user and not stored. The Staff table contains 100 rows, Branch table contains 5 rows, Clients contains 5000 rows, Viewing contains 20000 rows, and Property contains 500 rows. If there is any unknown information needed to make your calculation (for example, a statistic like the number of rows resulting from a selection must be invented in order to complete a calculation), clearly state the assumptions you make.

[10 marks]

[Total for Question 2: 30 marks]

3) [Transactions]

- (a) Is the following interleaving conflict serializable or not? Explain your answer. In addition, if it is conflict serializable, what is the equivalent serial schedule?

T1: Read(x)
T2: Read(x)
T2: Write(y)
T3: Read(y)
T1: Write(x)
T3: Write(y)

[10 marks]

- (b) Is deadlock possible for the following set of transactions? Explain why or why not.

T1: Lock_write(A) Read(A) Lock_write(B) Write(B) Write(A) Unlock(A) Unlock(B)	T2: Lock_write(B) Lock_write(C) Read(B) Read(C) Write(B) Write(C) Unlock(B) Unlock(C)	T3: Lock_write(C) Read(C) Write(C) Lock_write(A) Unlock(C) Write(A) Unlock(A)
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[10 marks]

- (c) Explain what the immediate update and deferred update storage schemes are, and how they affect recovery.

[5 marks]

[Total for Question 3: 25 marks]

- 4) [Distributed databases] Consider the following schema for a database that assists the function of a social media that has different "spaces" or communities that users congregate and post in:

Users (userID, username, email, profilePic, etc.)

Spaces (spaceName, description)

Posts (postID, creatorID {FK to Users}, spaceName, timestamp, contents)

Comments (commentID, posterID {FK to Users}, parentComment {FK to Comments}, timestamp, contents)

Moderation (spaceName, userID)

Membership (userID, spaceName)

The database is much too big to store on a single computer. Spaces tend to draw members from all over the world, though some spaces are geographically-specific. Propose a distribution scheme for this database and justify your choices. In particular, discuss how you would handle fragmentation, allocation, and replication, and mention consideration(s) you take into account in your proposal.

[Total for Question 4: 20 marks]

- 5) [Non-relational databases] Short answer questions (1-2 sentences).

- (a) Summarize the main differences between querying semistructured databases vs. querying relational databases.

[5 marks]

- (b) True or false: Hashing the keys of a table can be a quick way of figuring out how to fragment it. Explain your answer.

[5 marks]

[Total for Question 5: 10 marks]

END OF PAPER